

Engineering Curves: Construction of Cycloid

Rolling Circle Method with Tangent & Normal

Gajavada Sanjeevkumar

Problem Statement

Question

Draw a **cycloid** generated by a circle of diameter $D = 50$ mm rolling along a straight base line for **one complete revolution** without slipping.

Construct a **tangent** and a **normal** to the cycloid at any point P located on the curve.

Step 1: Draw Generating Circle & Directing Line

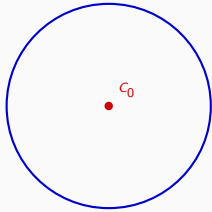
C_0 → Initial Center Point

AA' → Directing Line

Axis → Center Line



Step 1: Draw Generating Circle & Directing Line

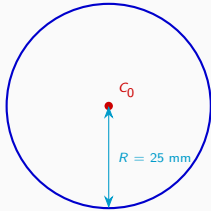


$C_0 \rightarrow$ Initial Center Point

$AA' \rightarrow$ Directing Line

Axis $\rightarrow$ Center Line

Step 1: Draw Generating Circle & Directing Line

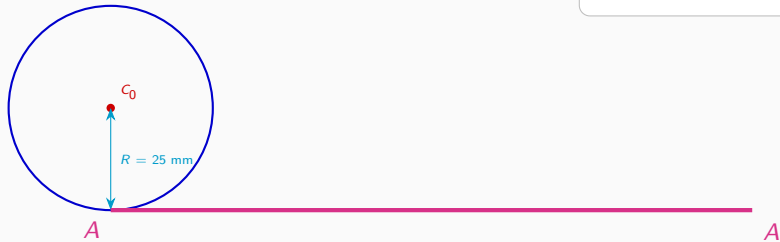


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Step 1: Draw Generating Circle & Directing Line



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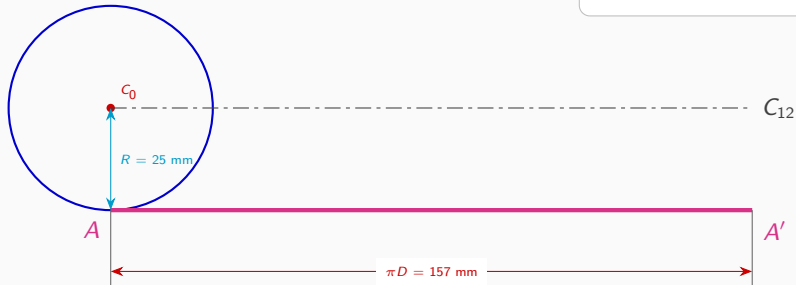


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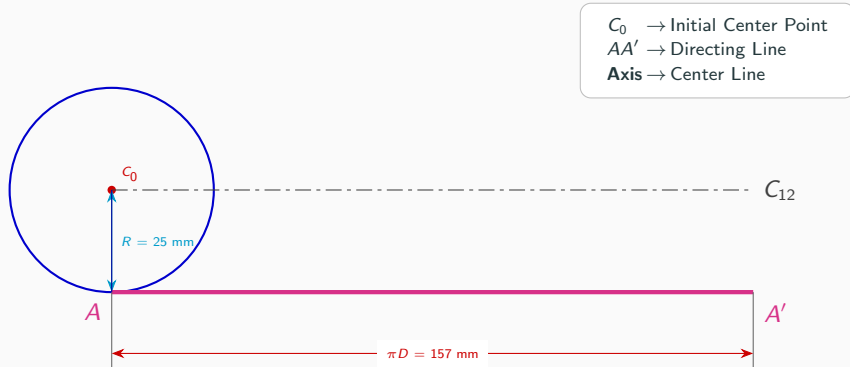


$C_0 \rightarrow$ Initial Center Point

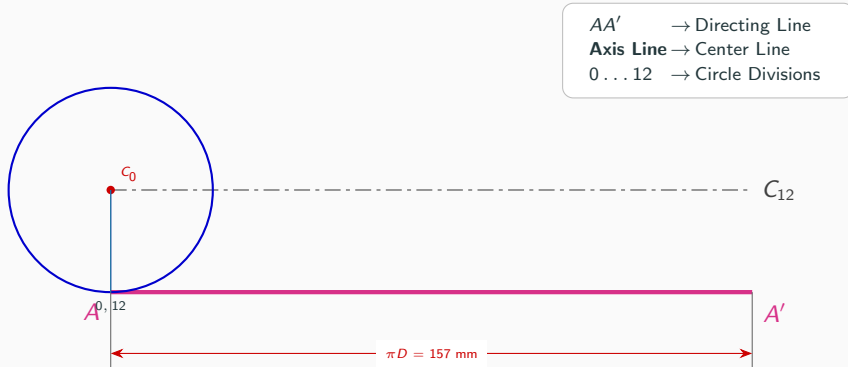
$AA' \rightarrow$ Directing Line

Axis $\rightarrow$ Center Line

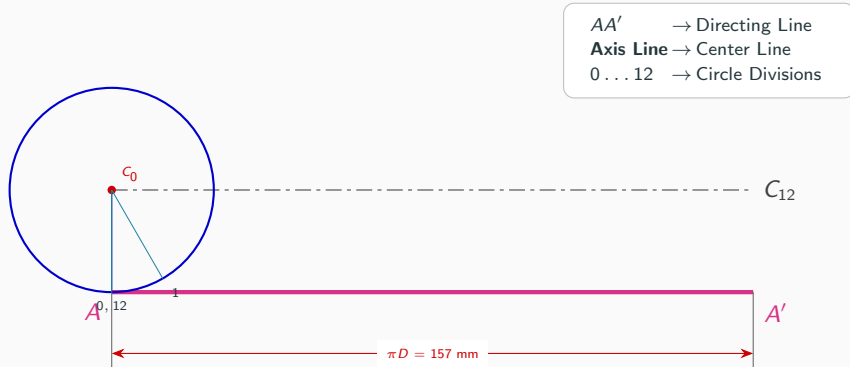
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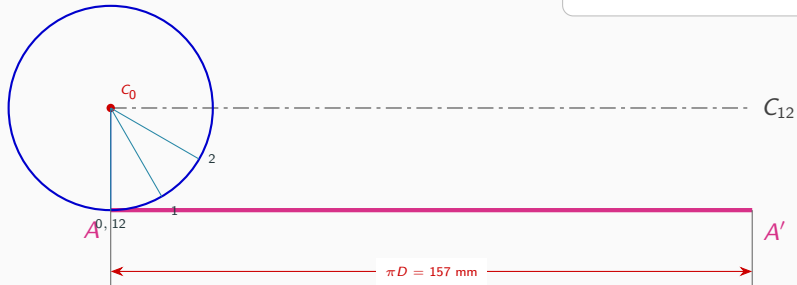
Step 2: Divide Generating Circle into 12 Equal Parts



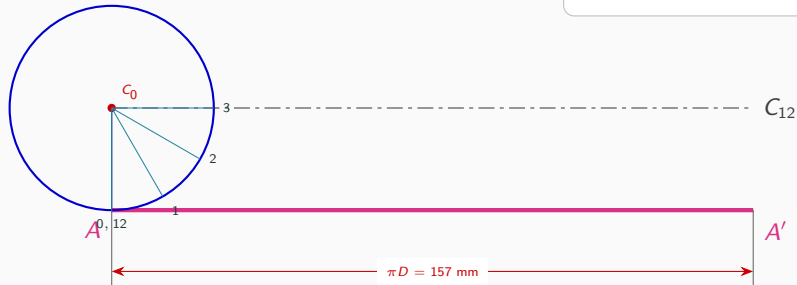
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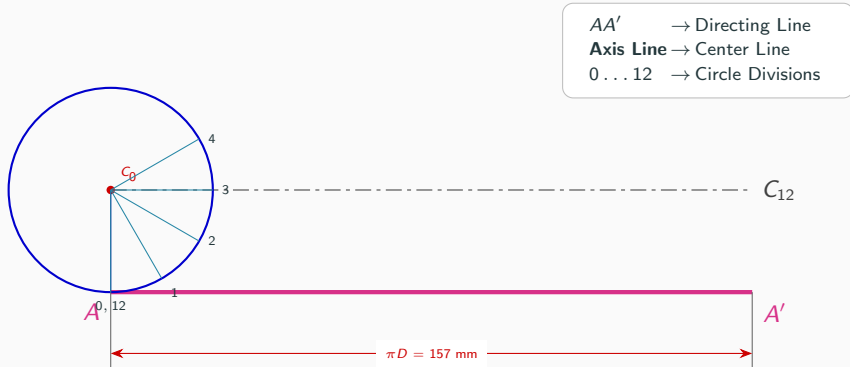


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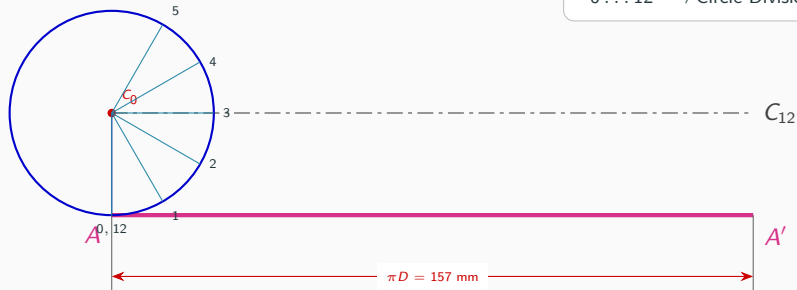


AA' → Directing Line
Axis Line → Center Line
 $0 \dots 12$ → Circle Divisions

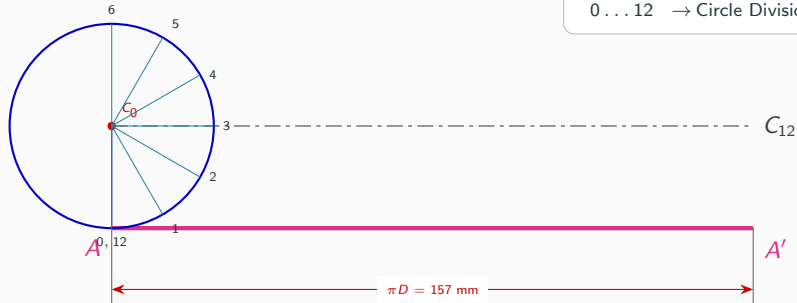
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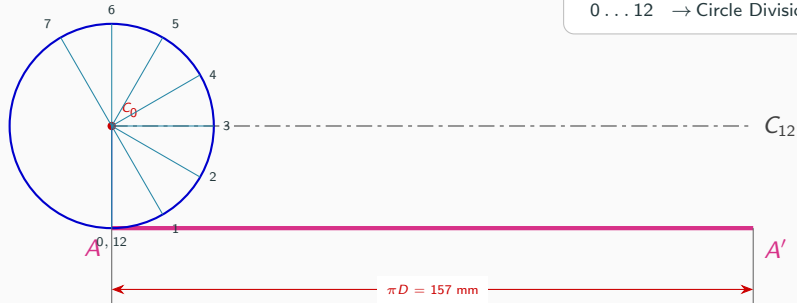
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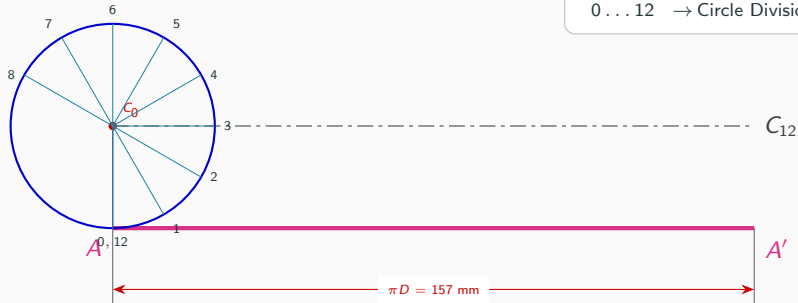
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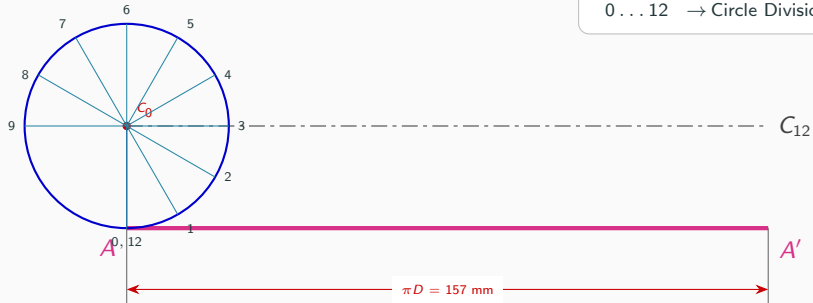
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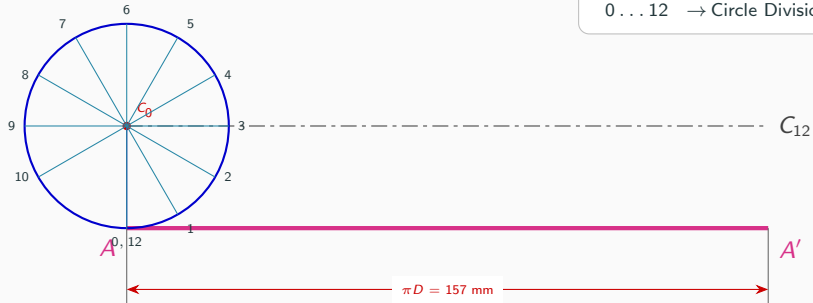
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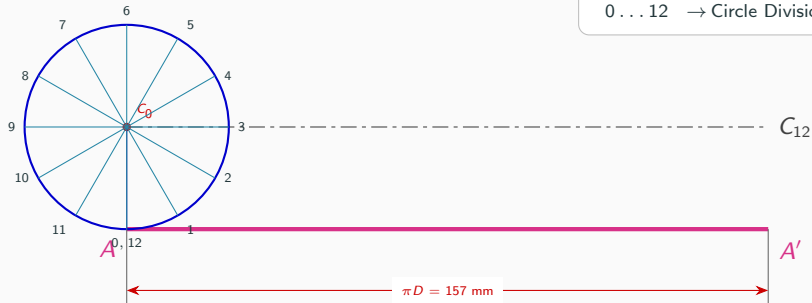
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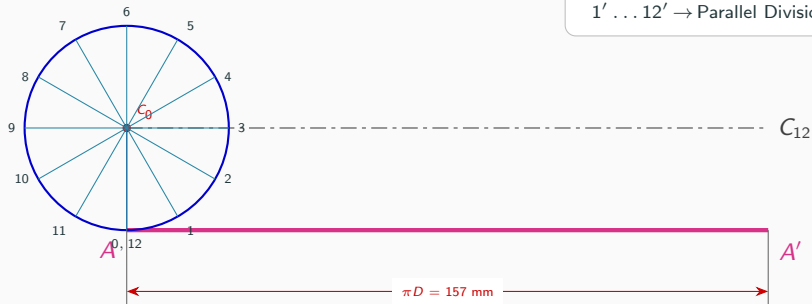
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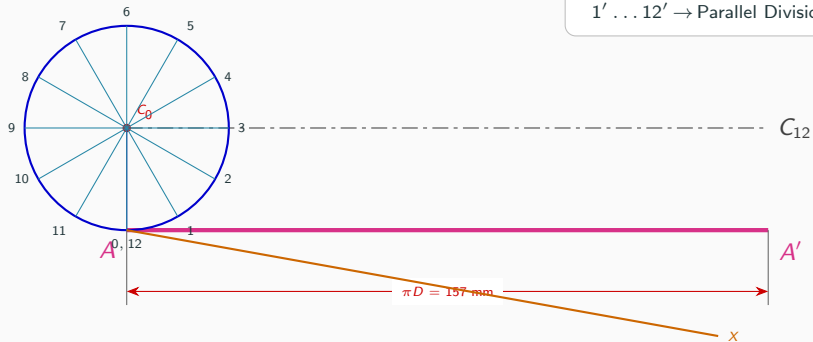
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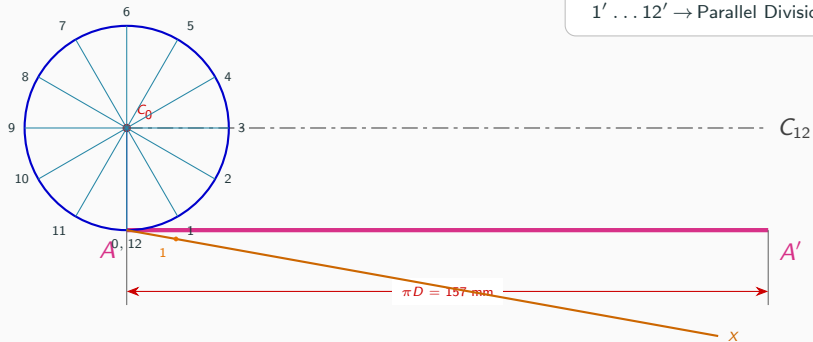
Step 3: Divide Directing Line using Line Division Method



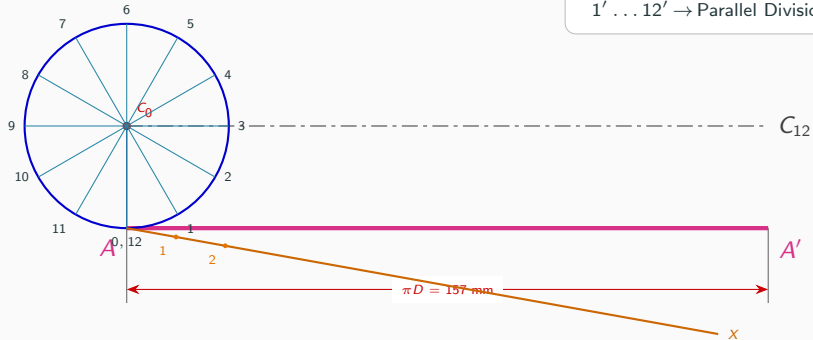
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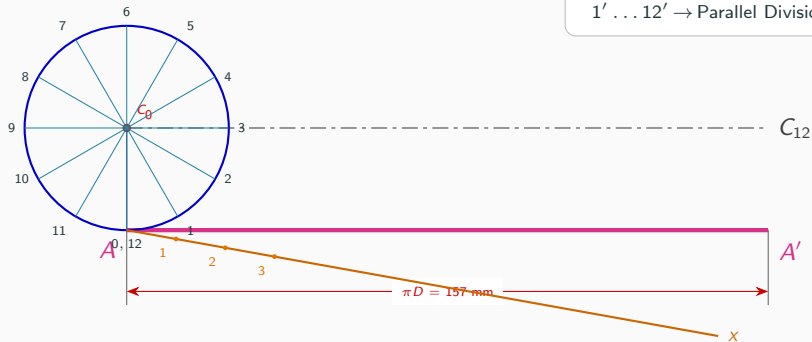
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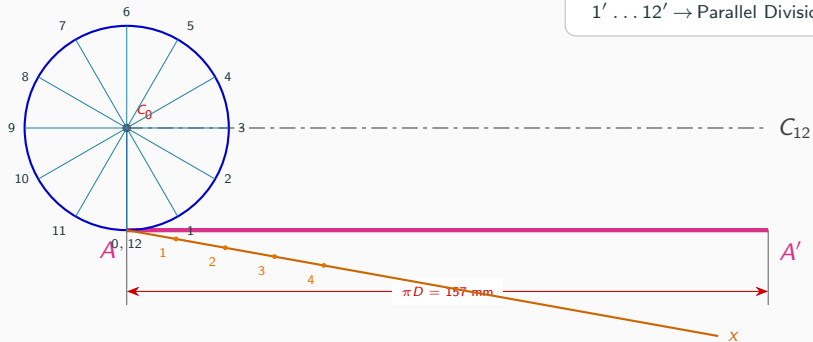
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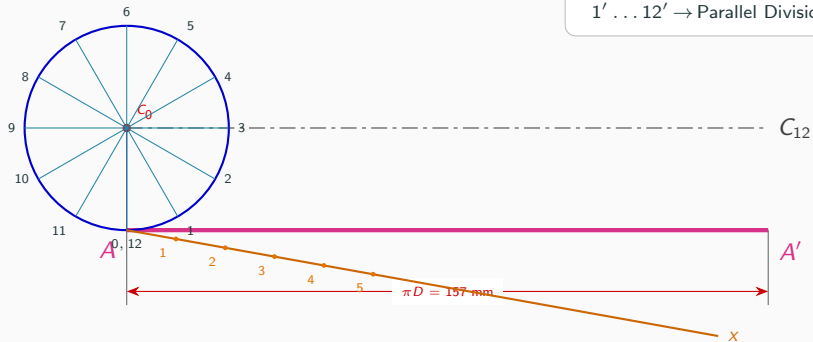
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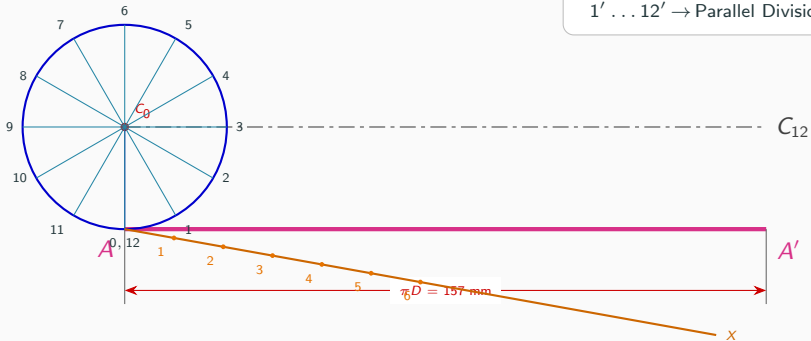
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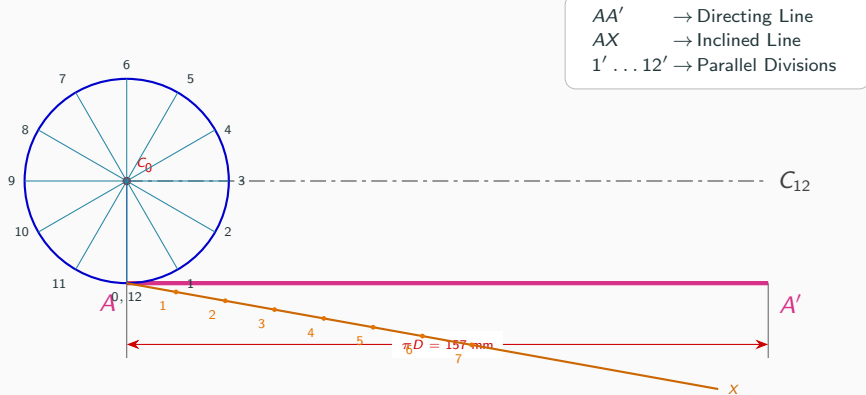


AA' → Directing Line

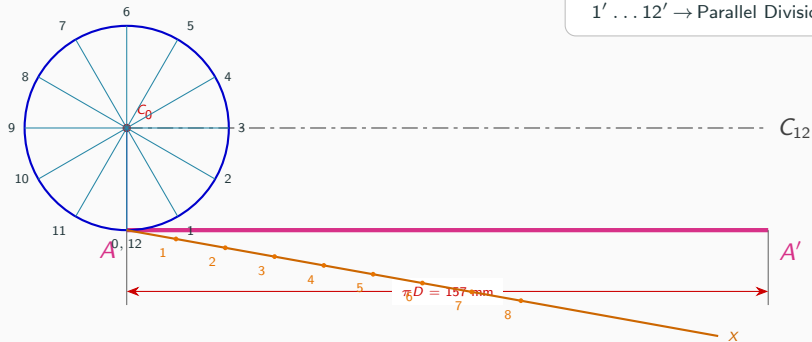
$AX \rightarrow$ Inclined Line

$1' \dots 12' \rightarrow$ Parallel Divisions

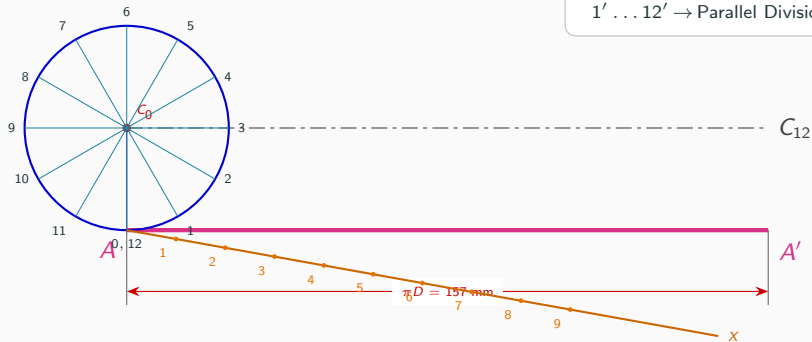
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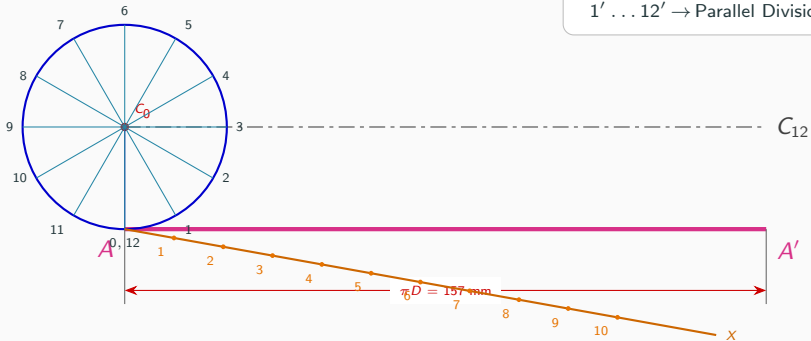
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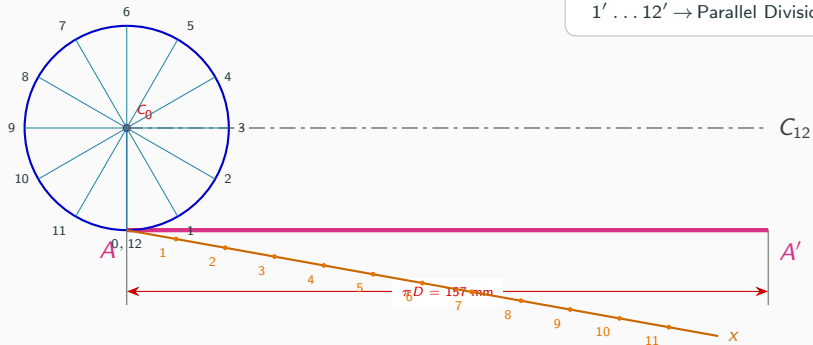


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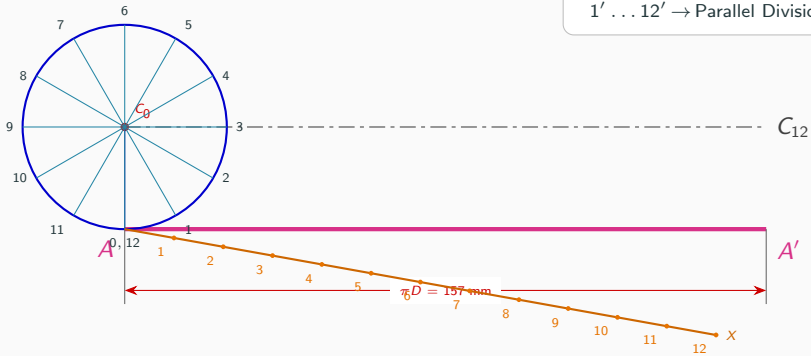


AA' → Directing Line
 AX → Inclined Line
 $1' \dots 12'$ → Parallel Divisions

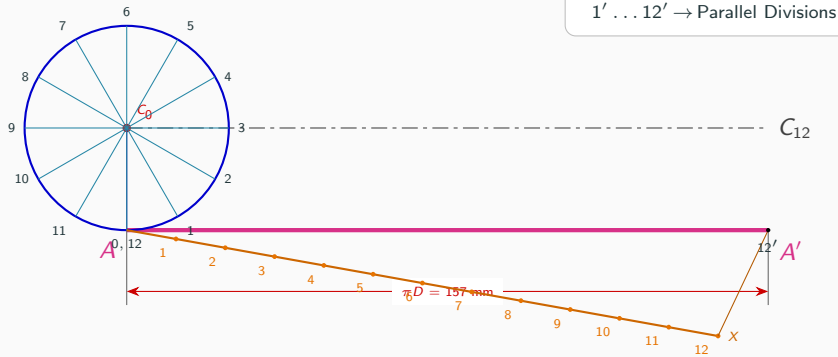
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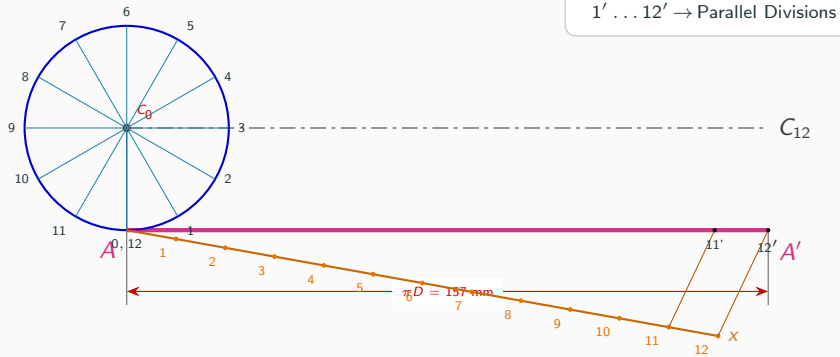
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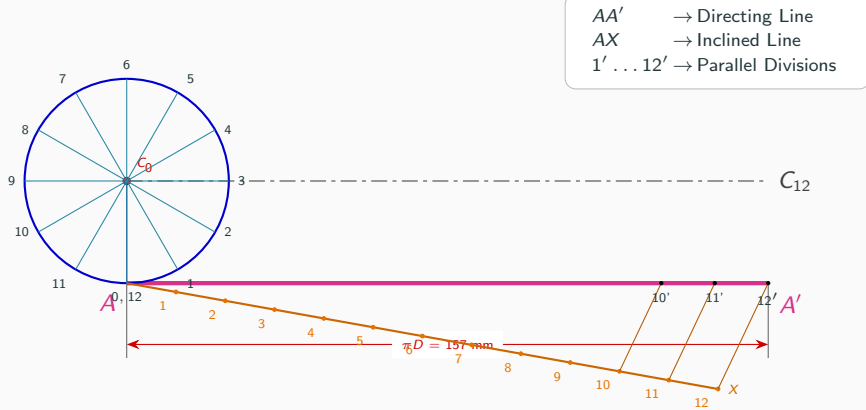
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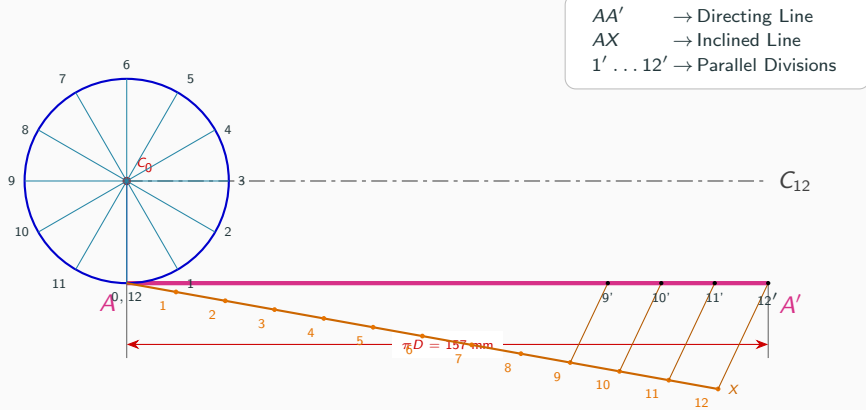
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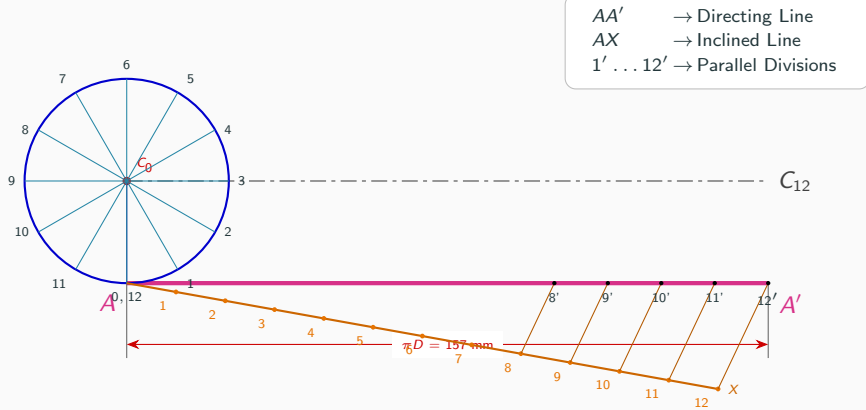
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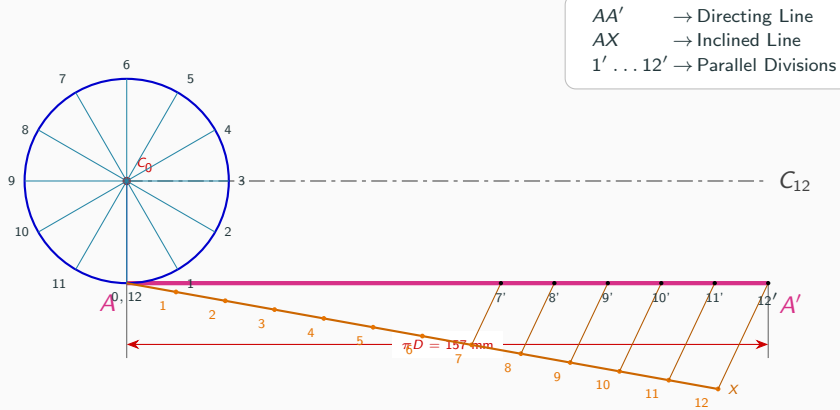
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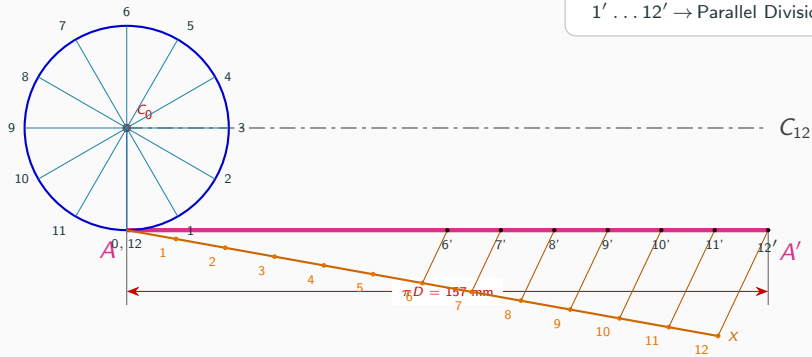
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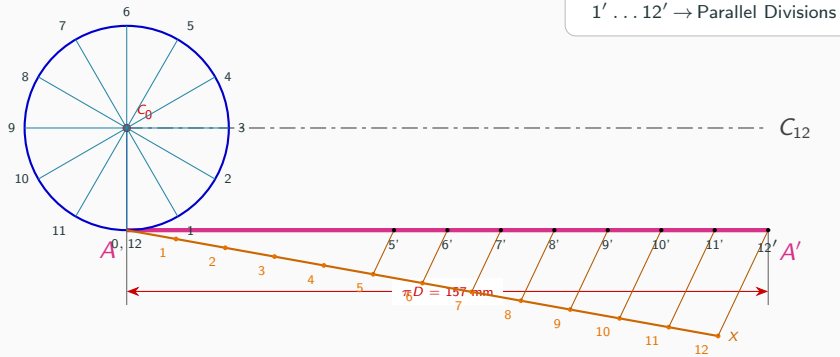
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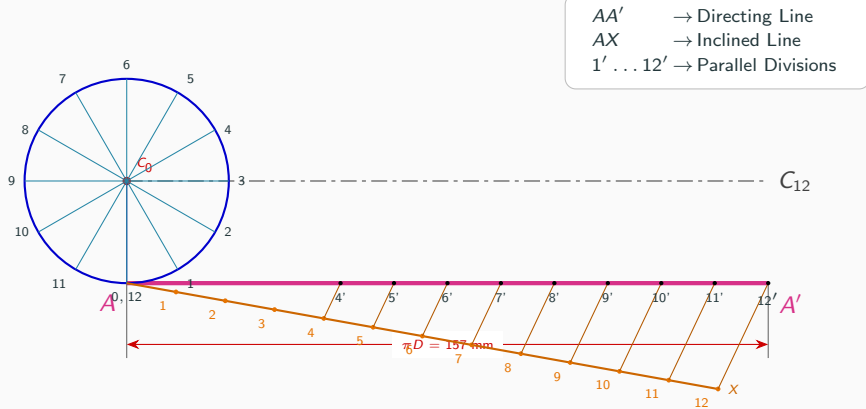
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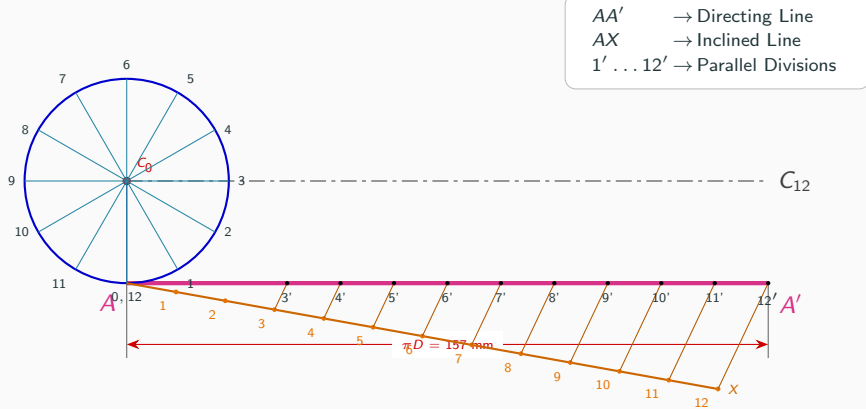
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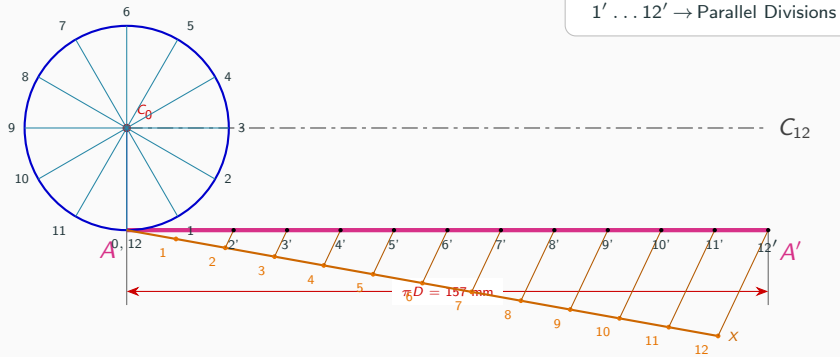
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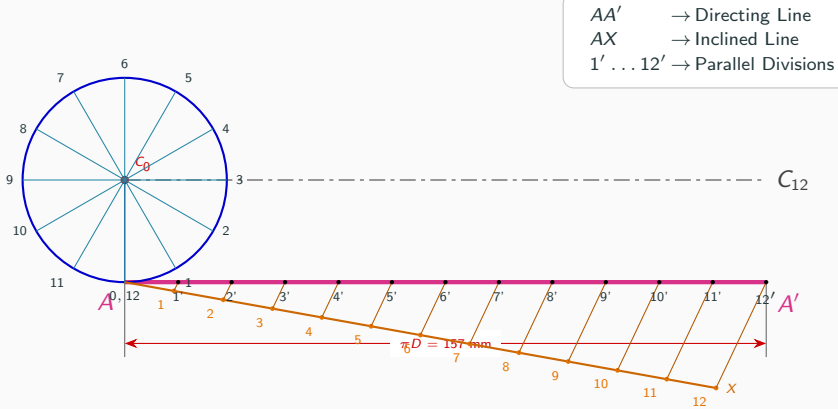
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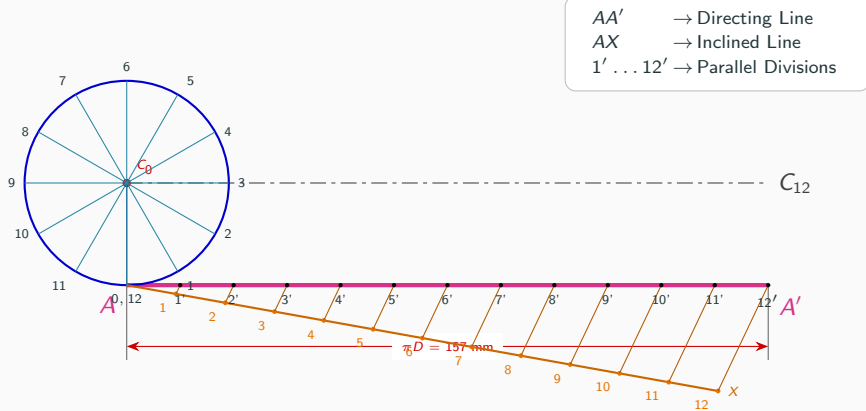
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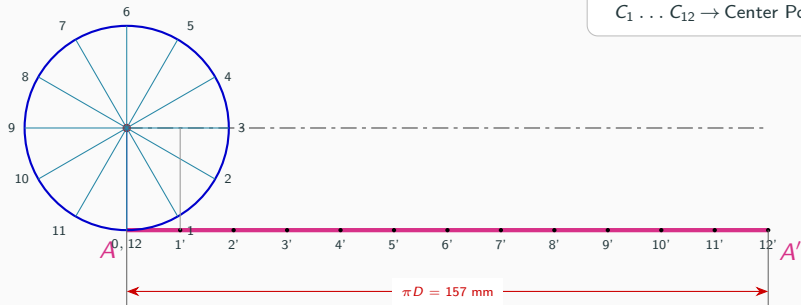
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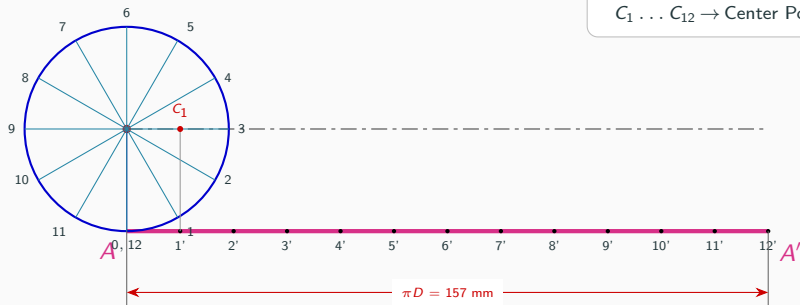
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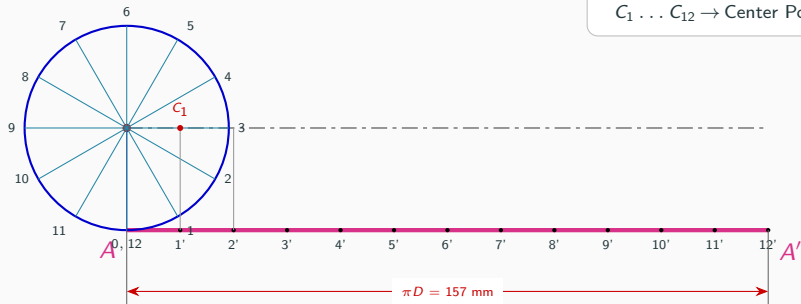
Step 4: Project Divisions to Locate Centers $C_1 \dots C_{12}$



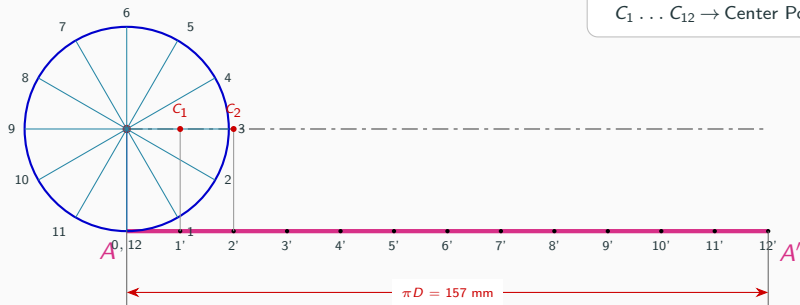
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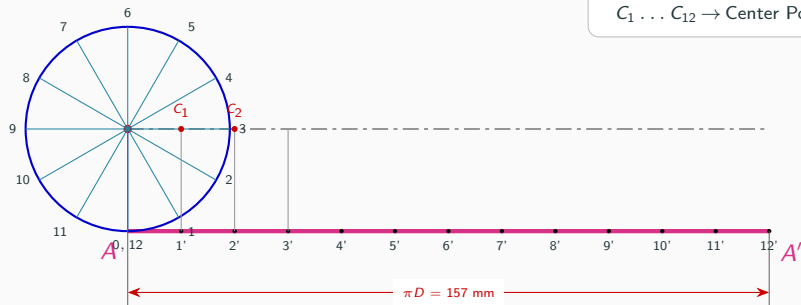
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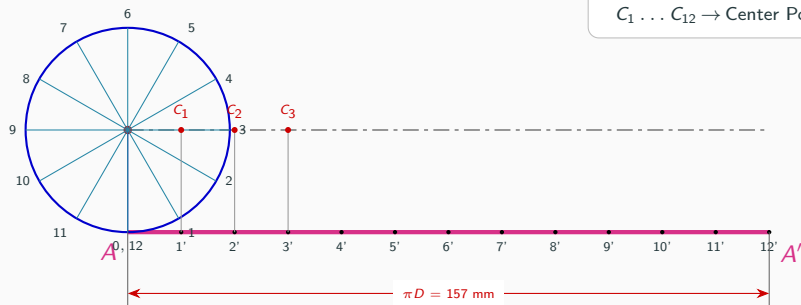
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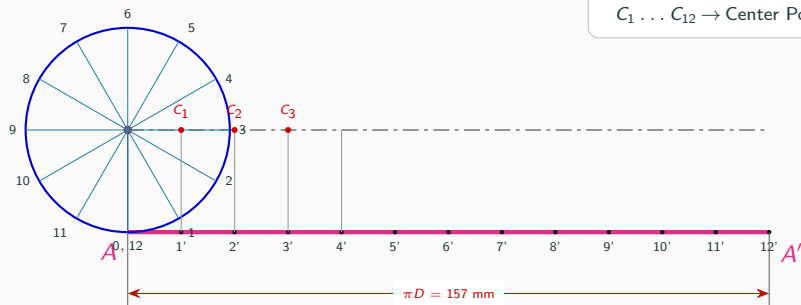
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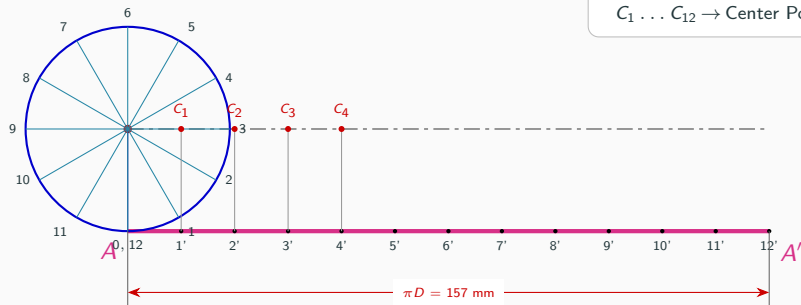
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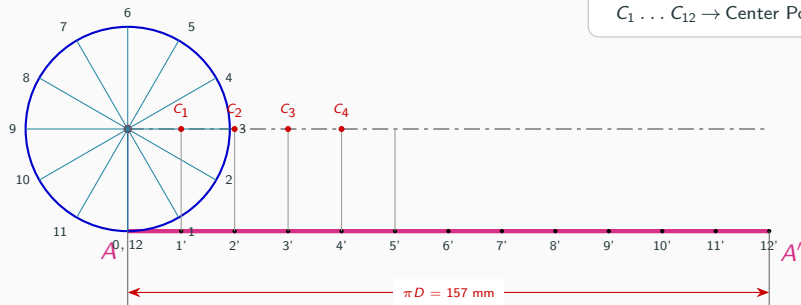
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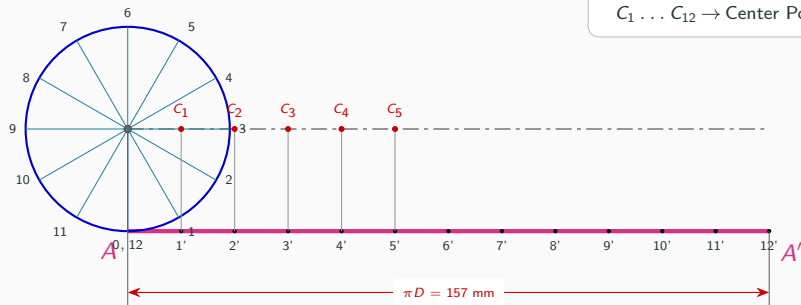
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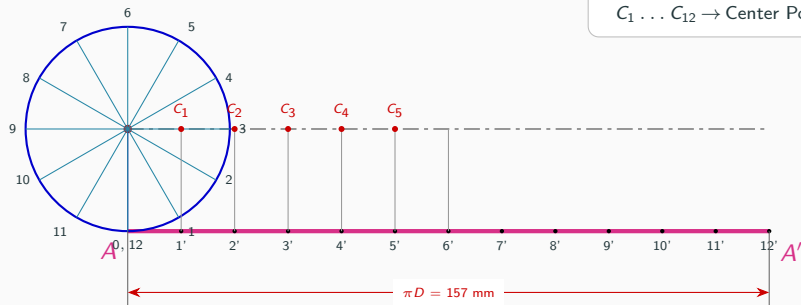
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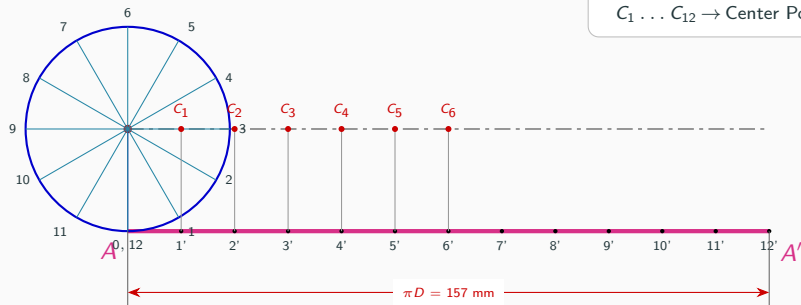
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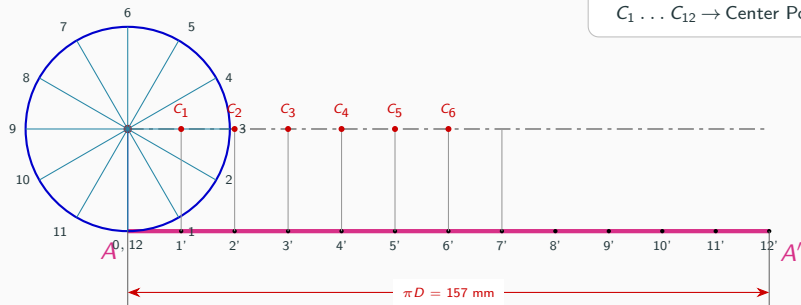
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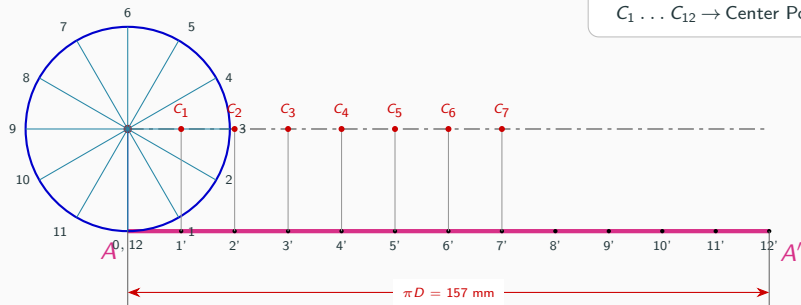
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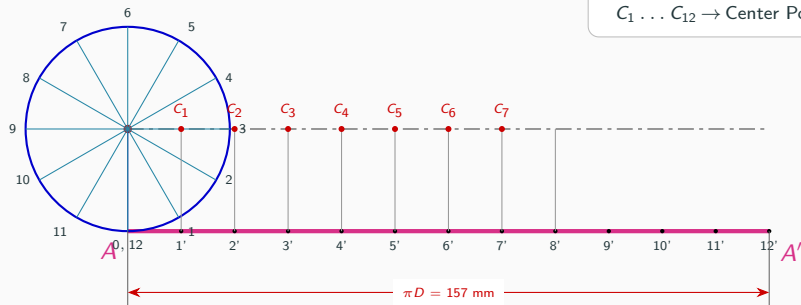
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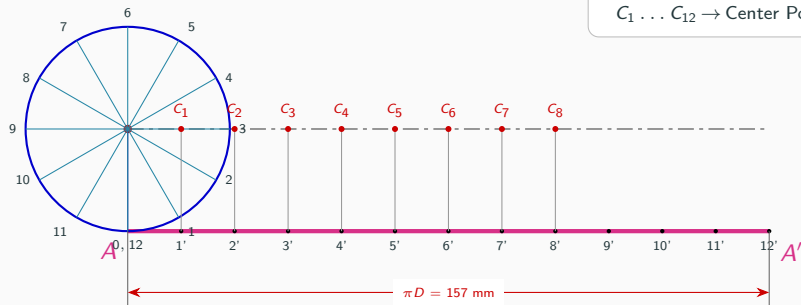
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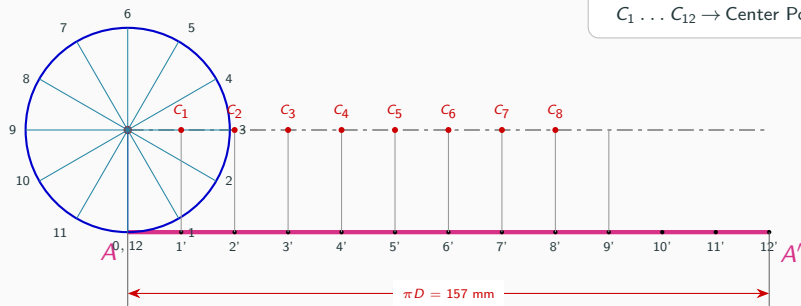
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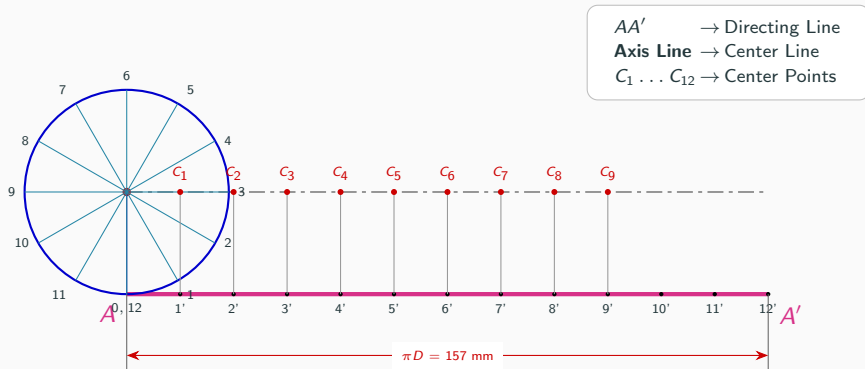
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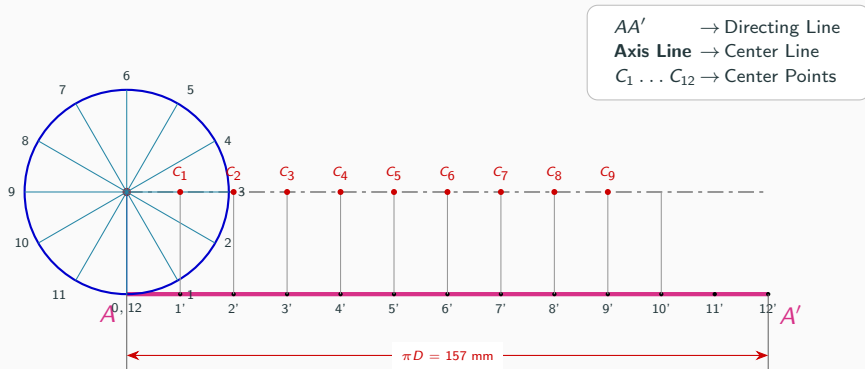
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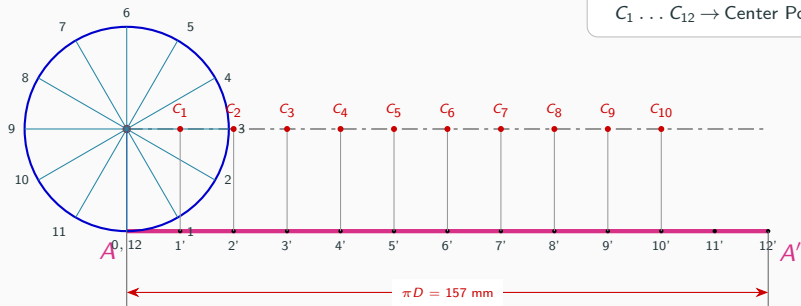
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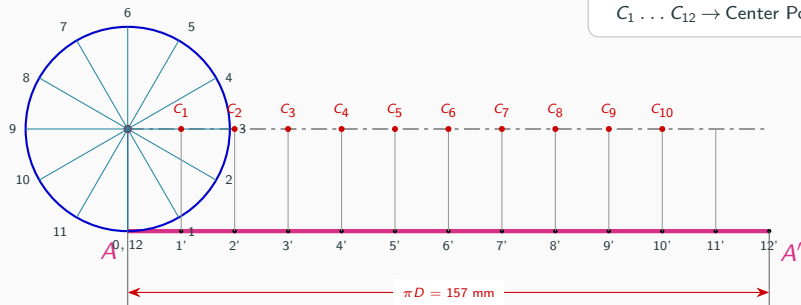


AA' → Directing Line

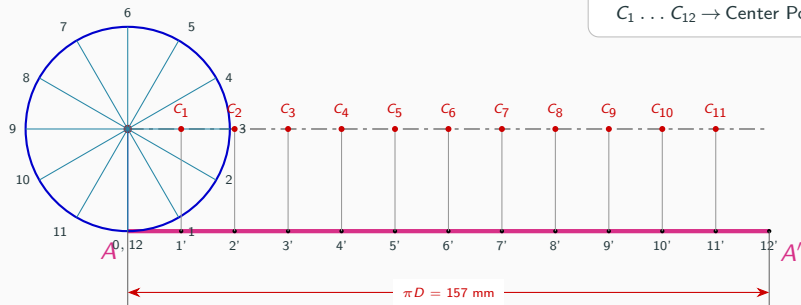
Axis Line → **Center Line**

$C_1 \dots C_{12} \rightarrow$ Center Points

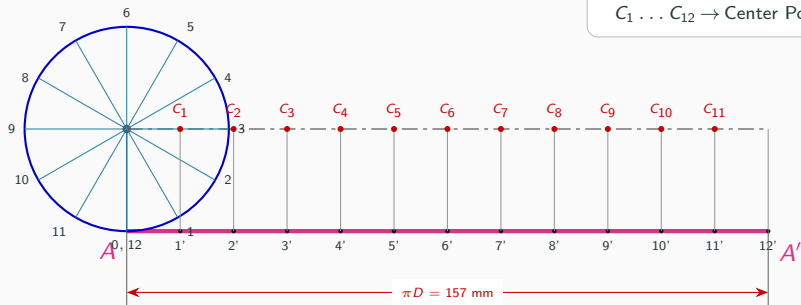
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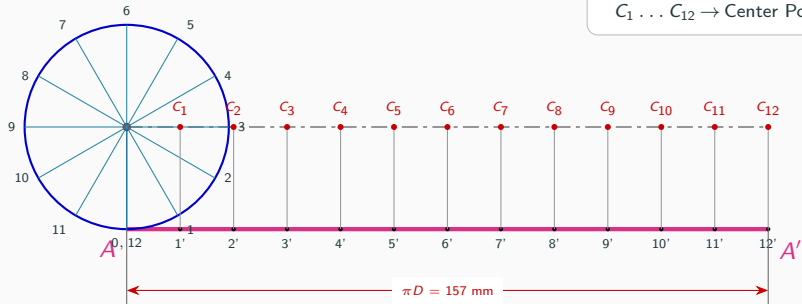


AA' → Directing Line

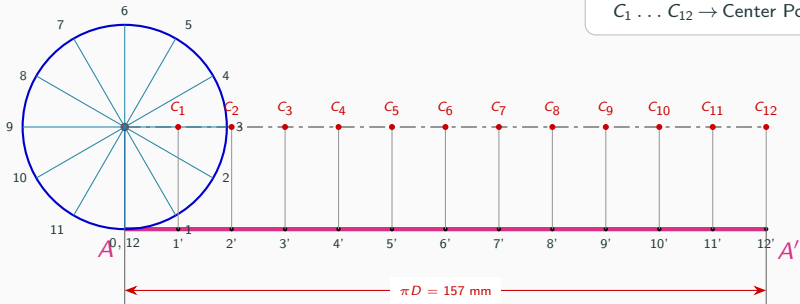
Axis Line → **Center Line**

$C_1 \dots C_{12} \rightarrow$ Center Points

Step 4: Project Divisions to Locate Centers $C_1 \dots C_{12}$



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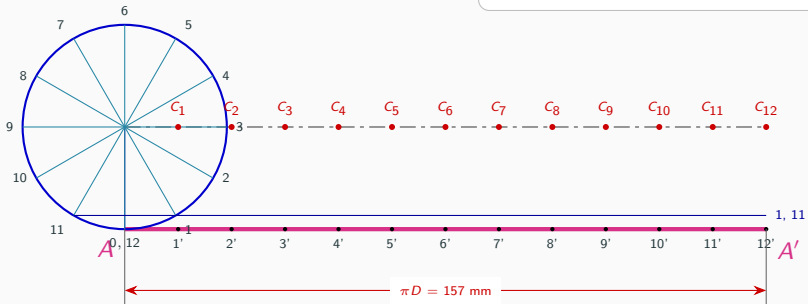


AA' → Directing Line

Axis Line → **Center Line**

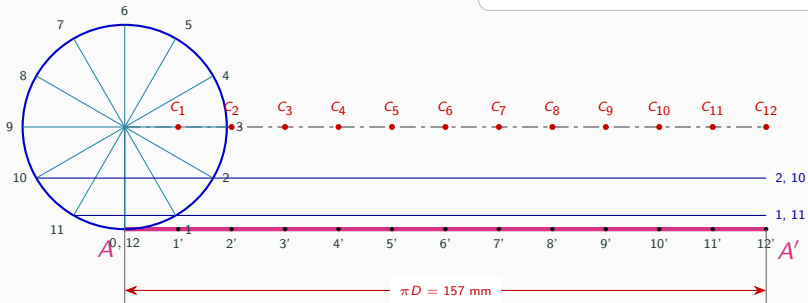
 $C_1 \dots C_{12} \rightarrow \text{Center Points}$

Step 5: Draw Horizontal Reference Lines



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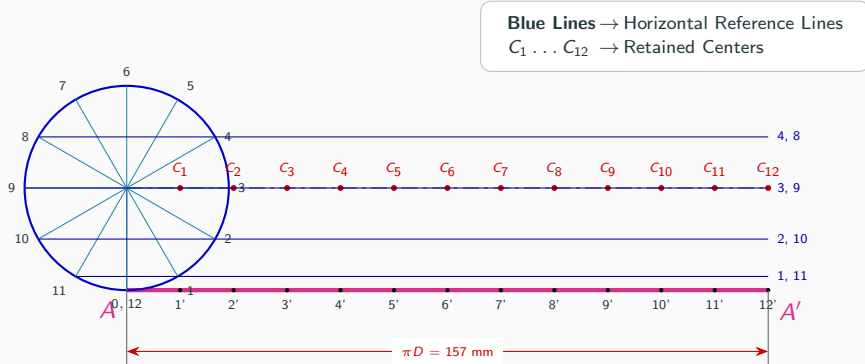
Blue Lines $\rightarrow$ Horizontal Reference Lines
 $C_1 \dots C_{12} \rightarrow$ Retained Centers



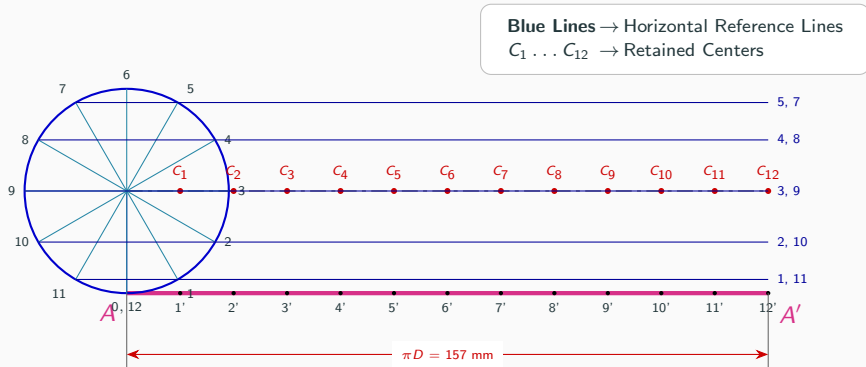
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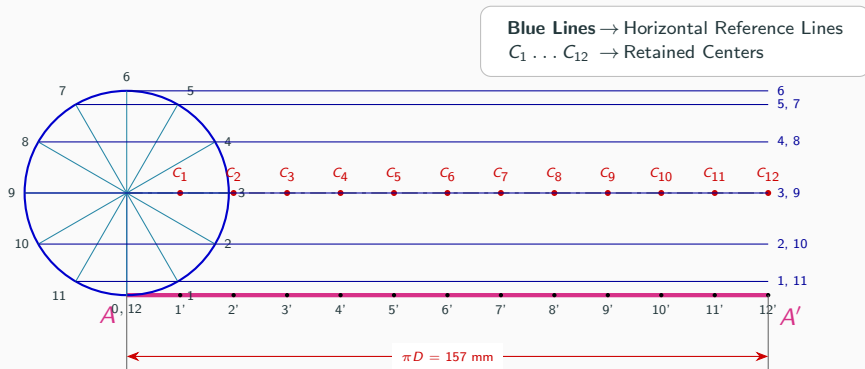
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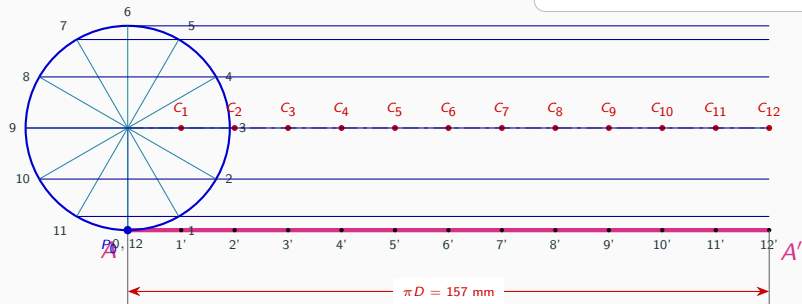
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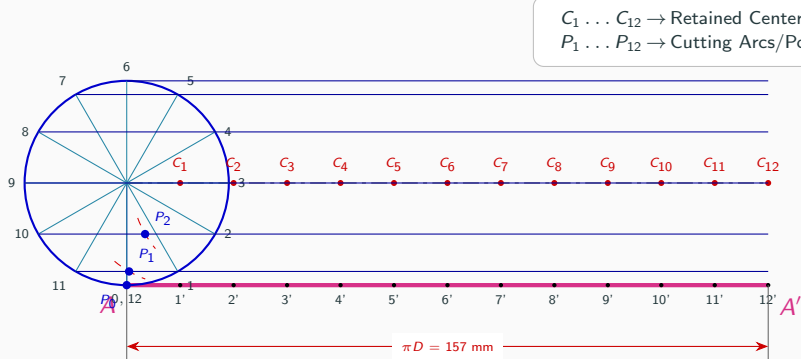
Step 5: Draw Horizontal Reference Lines



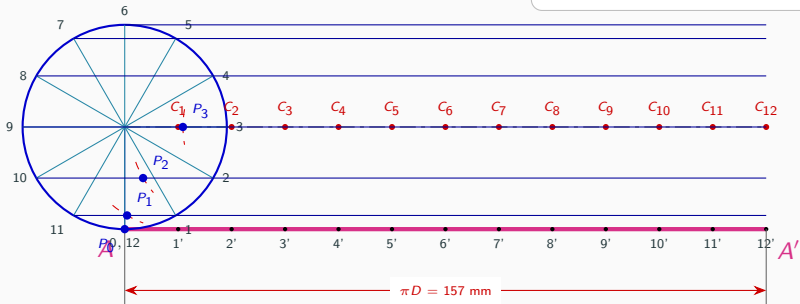
Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$



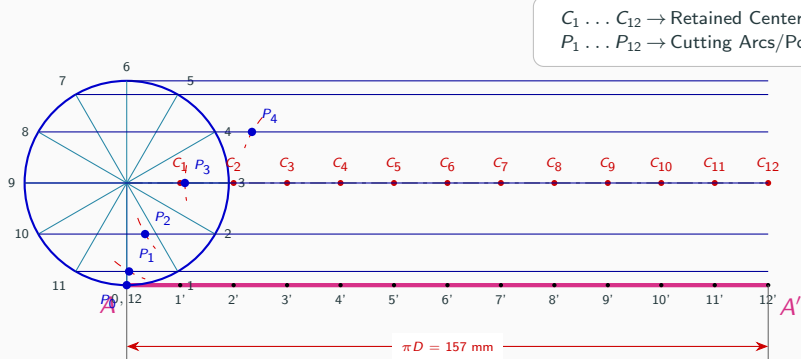
Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$



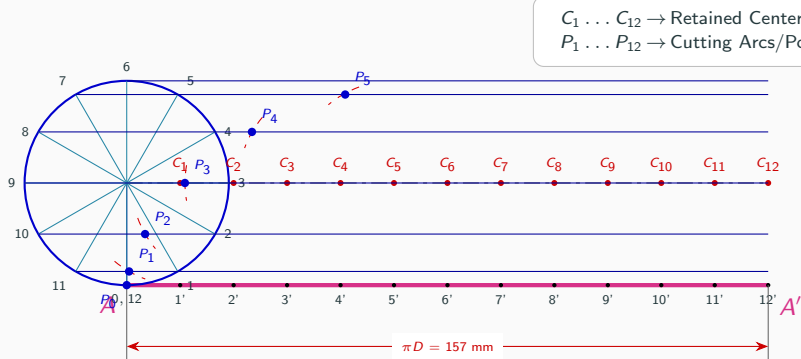
Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$



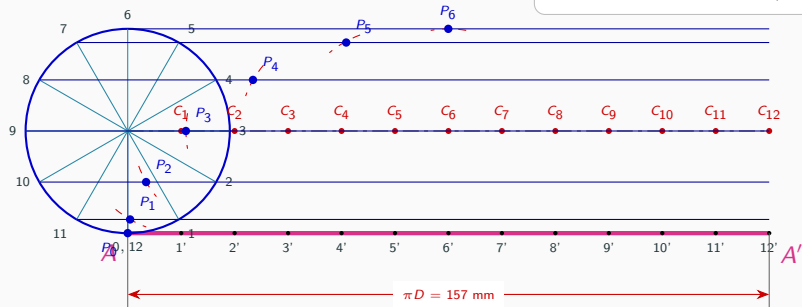
Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$



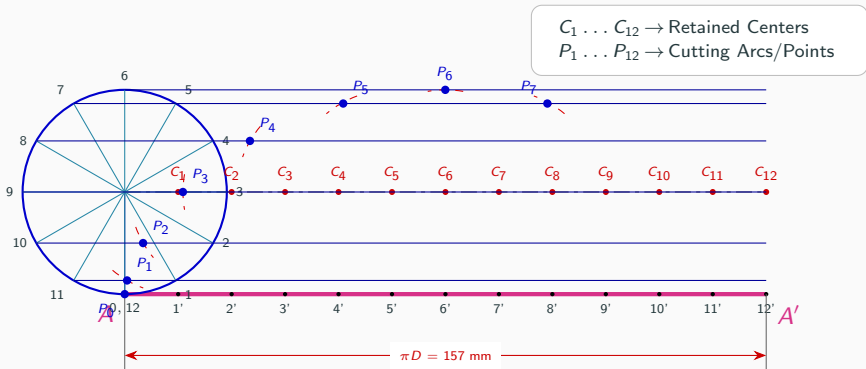
Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$



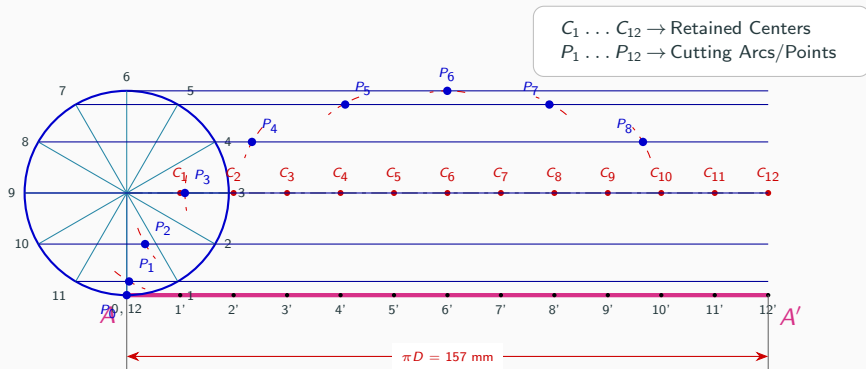
Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$



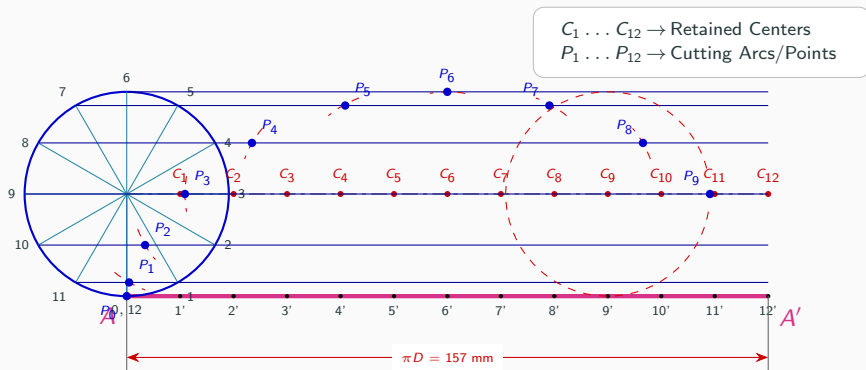
Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$



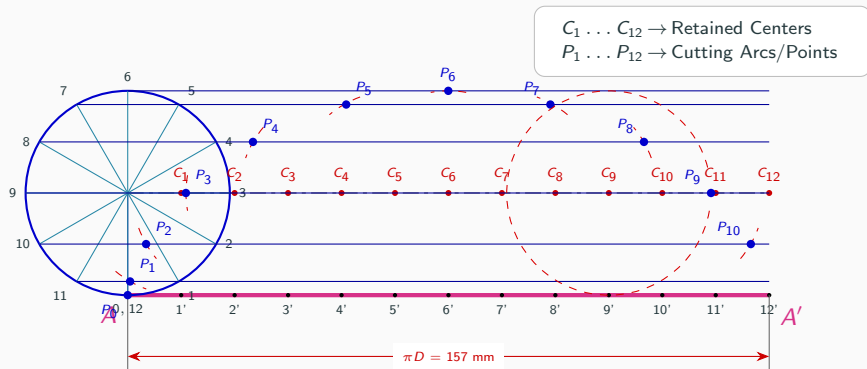
Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$



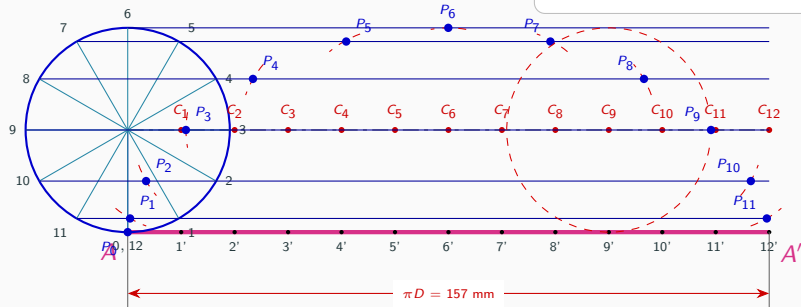
Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$



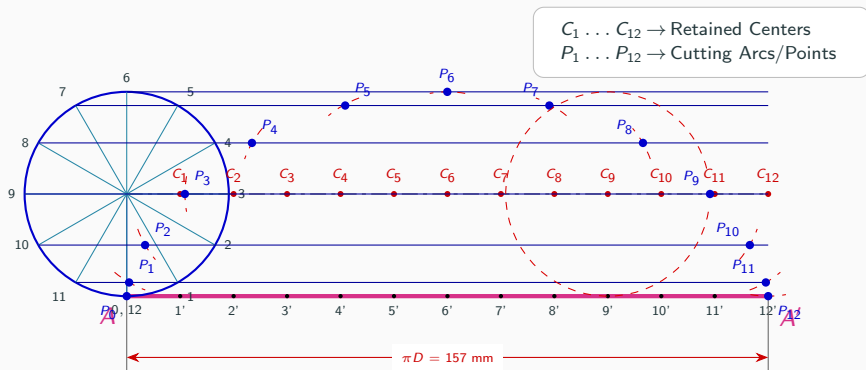
Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$



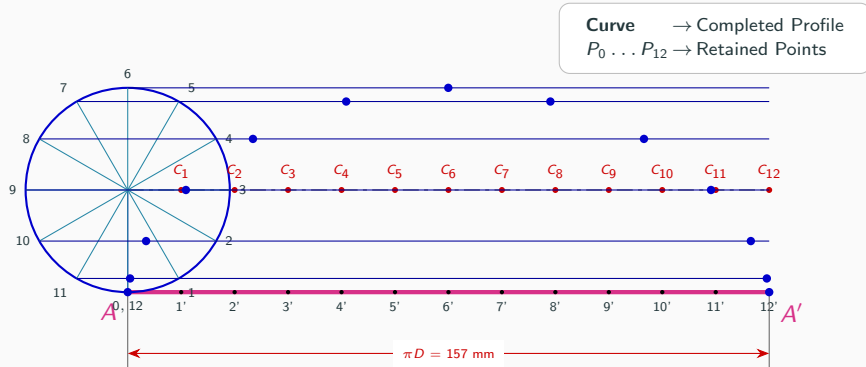
Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$



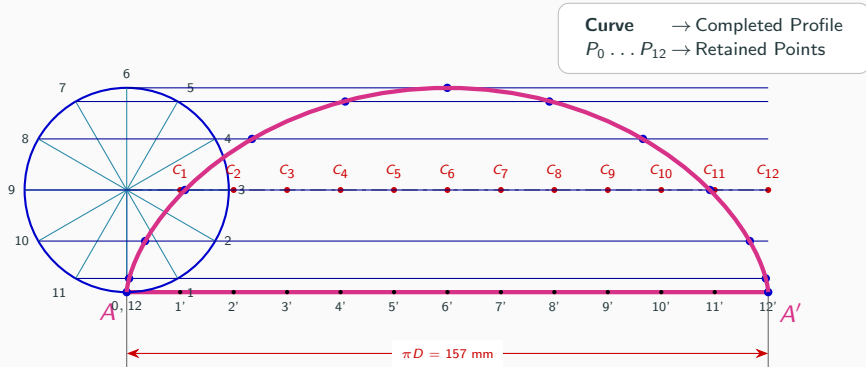
Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$



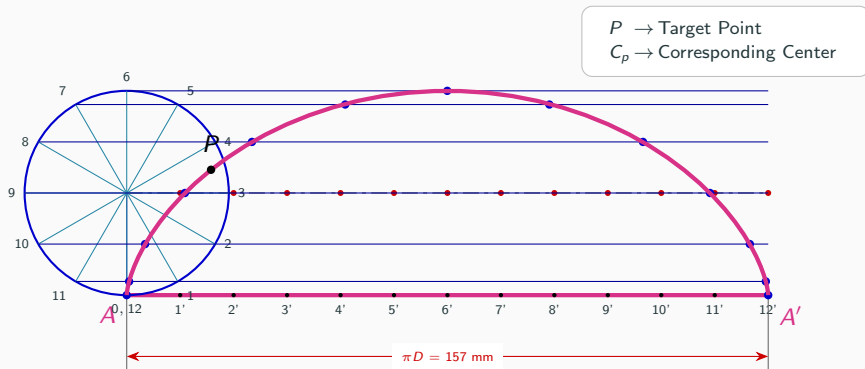
Step 7: Smoothly Join Points to Form Cycloid Curve



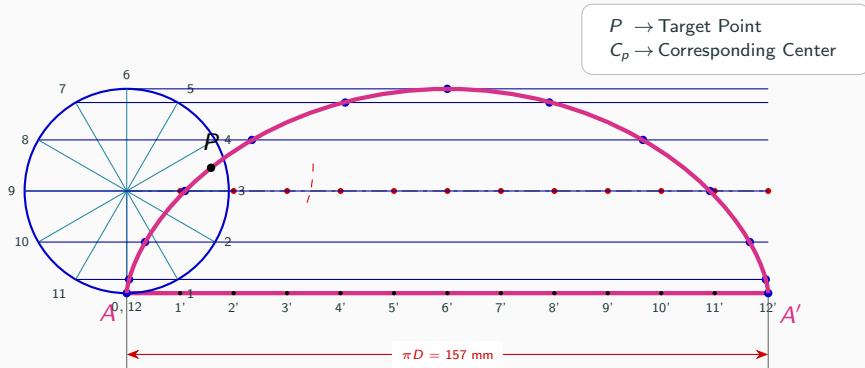
Step 7: Smoothly Join Points to Form Cycloid Curve



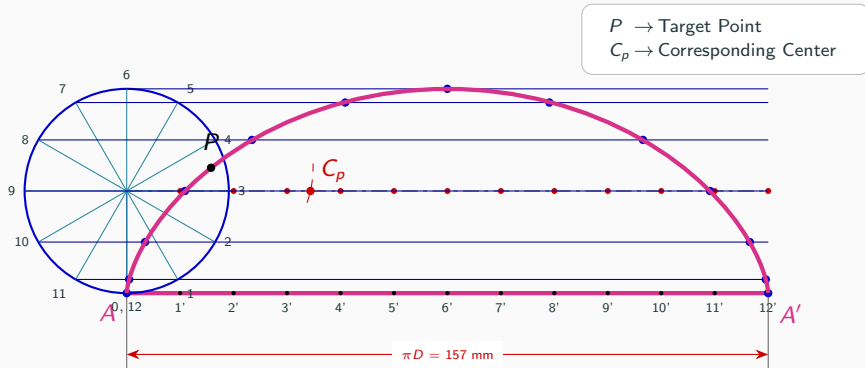
Step 8: Mark Point P & Locate Center C_p



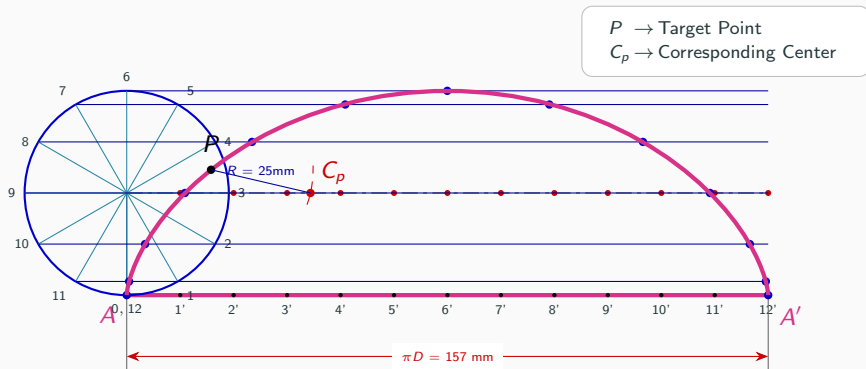
Step 8: Mark Point P & Locate Center C_p



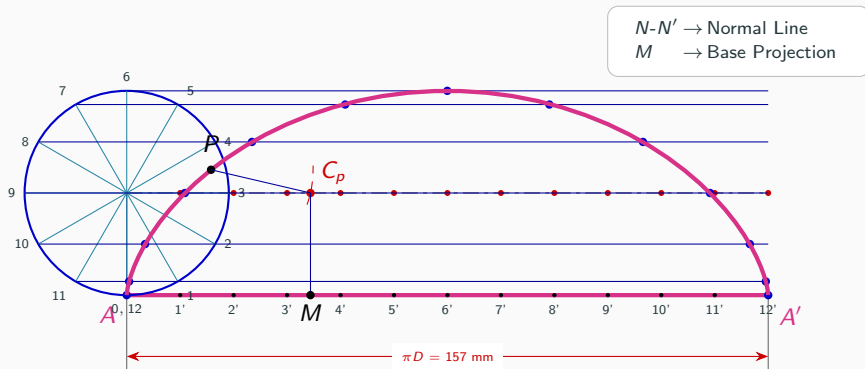
Step 8: Mark Point P & Locate Center C_p



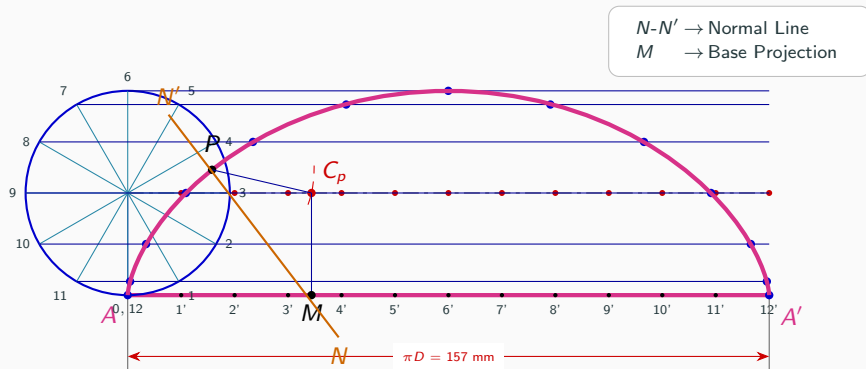
Step 8: Mark Point P & Locate Center C_p



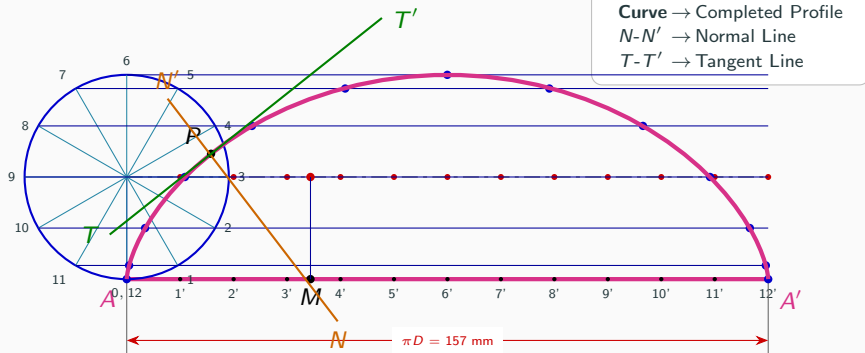
Step 9: Construct Normal Line $N-N'$



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Step 10: Construct Tangent Line $T-T'$ & Final Result



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